Explainable AI Clinical Decision Support System

Clinical Prediction Report

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Patient Information

Field	Value
Patient ID	PAT003
Patient Name	Wilson
Doctor	Dr. Vinoth
Hospital	AIIMS Hospital
Disease	Stroke
Age	63
Gender	Male
Hypertension	No
Heart Disease	No
Ever Married	Yes
Work Type	Private
Residence Type	Urban
Average Glucose Level	210.45
Body Mass Index (BMI)	31.2
Smoking Status	never smoked

Prediction Result

Field	Result
Prediction	No Stroke
Probability	63.55 %
Risk Level	Low

Prediction Class	0

Clinical Recommendation

No significant indication of stroke was detected by the model. Maintain a healthy lifestyle, monitor blood pressure and blood sugar, exercise regularly, and continue routine health check-ups.

Top Influencing Features (SHAP)

Feature	Impact
Age	-3.7841
Average Glucose Level	-0.4746
Body Mass Index	-0.1882
Work Type - Children	-0.1588
Smoking Status - Never Smoked	-0.1224

LIME Explanation

Local Interpretable Model-Agnostic Explanation (LIME) showing the most influential conditions used by the model for this prediction.

Feature	Impact
age > 0.78	0.5549
work_type_children <= -0.40	-0.1152
hypertension <= -0.33	-0.0816
avg_glucose_level > 0.17	0.0575
work_type_Self-employed <= -0.44	0.0563

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • Flask • React

Doctor Signature: _____

Patient Signature: _____